

Feature pyramid network with multi-scale prediction fusion for real-time semantic segmentation [☆]

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ABSTRACT

Feature pyramid network (FPN) is constructed from a bottom-up pathway and a top-down pathway. The method involves multi-scale features, so it can obtain rich contextual information from lower scales and high resolution from the largest scale. Additionally, different receptive fields are effective to capture both thin and large objects in image scenes. All feature maps concatenate together to predict the targets. However, the average pooling method yields the problem of combining the best predictions with poorer ones. In this paper, we proposed a dual prediction to leverage the useful characteristics of each FPN feature map. A low scale prediction attains good precision for large objects. The other one suitably segments narrow objects. Finally, a multi-scale fusion is deployed with an attention part. The attention module finds pixels of a low scale having high probabilities of wrong labels, and then requires the supplements from a high scale. A multi-scale fusion allows the network to learn across the different scales of predictions. We have achieved good Results 77.9% mIoU at 62 FPS on Cityscapes and 44.1% mIoU on Mapillary Vistas.

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1. Introduction

Alongside the innovation of technologies such as fast computers or servers and various types of modern cameras, semantic segmentation of the computer vision field has been concentrated to create new applications. This method is utilized to understand the image scenes at the pixel levels, where each pixel will be assigned to corresponding labels of classes. This technology has wide applications in autonomous driving, robot vision, and public safety. Although many approaches of semantic segmentation have been proposed, they still suffer the limitations of accuracy segmentation and inference speed.

For self-driving applications, they demand both rapid speed and high accuracy from the model. Early, they designed lightweight structures to obtain the real-time semantic segmentation such as

joining features of both local features and surrounding context [1], building a bottleneck by combining depth-wise asymmetric convolution and dilated convolution [2], or grouping point-wise and depth-wise dilated separable convolutions [3]. On the other hand, many approaches improve the network efficiency by using residual connections and factorized convolutions [4], creating a multispectral image segmentation with pixel-level annotation [5], and deploying grouped convolution and channel shuffling [6]. The time consumption of the networks is significantly reduced, but accuracy is not suitable for autonomous applications. In order to improve the accuracy, they propose new algorithms for semantic segmentation such as a prototype view [7] or the cross-image pixel contrast [8]. Prior approaches only use a top layer to predict the output results, but this layer is hard to recover the spatial detail information. Some studies employ feature fusion from multiple branches to handle spatial loss [9,10].

In street scenes, the objects have diverse sizes, so it is difficult to cover a long range of sizes with a single receptive field. The drawback leads to low segmentation accuracy. DeeplabV3 [11] designed dilated convolutions with flexible rates and multiple fields of view to capture different contextual information. Feature pyramid network (FPN) is a useful architecture for the computer vision field when it contributes to the feature pyramid to generate final outputs. This method commonly applies not only for object detection

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[12] but also for semantic segmentation [13]. In Fig. 1a, it is a FPN proposal for semantic segmentation [13]. The FPN features include multiple scales extracted from ResNet backbone and then are concatenated together to predict the results. The concatenated layer mainly recovers rich semantic features from lower scales and captures spatial information from the highest scale. Hence, It obtains high resolution and the different receptive fields from multi-scale features. Other approaches relate to feature pyramids, but they upgrade the decoder part [14,15]. In Fig. 1b, this is one case of decoder modification [16]. Instead of using normal convolutions and concatenation functions, they deploy a multi-scale attention pyramid to process features extracted by ResNet network. The attention module further improves the robustness of various scale distributions. Then, they utilize adaptive fusion for multiple prediction results. The FPN decoder modification has significantly improved the aspect of accuracy segmentation, but it is not real-time semantic segmentation.

In this study, we continuously concentrate on FPN decoder modification. FPN backbone is inspired from baseline method [13]. We propose a dual prediction module to decode feature instead of processing with concatenation and predicting by average pooling. This module processes two pairs of features to generate segmentation separately. In this way, we can effectively capture the characteristics of each FPN feature. Multi-scale fusion is utilized to combine two-scale segmentation. We deploy an attention module, which determines the low-precision areas of a low-scale segmentation and then selects learned weights from a high-scale segmentation to supplement.

Our main contributions can be summarized as follows:

- A dual prediction is proposed to suitably capture both the thin and large objects from complex street scenes.
- A multi-scale fusion method is designed to effectively combine prediction from two scales.
- We have achieve the good Results 77.9% mIoU at 62 FPS on Cityscapes and 44.1 mIoU on Mapillary Vistas.

2. Related works

Multi-scale context methods: In semantic segmentation, the rich spatial information provides a fine detail for objects, but the receptive field should be shrunken. The shrunken receptive field will have challenge to cover the whole context of large objects. Intuitive approaches designed the network to capture disparate context information from different receptive fields. In [11,17], context information is collected from multiple dilated convolutions with flexible rates. PSPNet [18] utilized a pyramid pooling module in which information is obtained from different scales and varying

among different sub-regions. In [19], a pyramid spatial pooling adds the atrous convolution with different dilation rates to extract dense feature maps. Dynamic multi-scale filters are employed to handle a specific scale related to input [20]. Previous studies mainly capture information by various receptive fields at the final layer of encoder. In convolution neural network (CNN), the first layers will contain more detail information for object's boundaries. Many methods attempt to gather disparate receptive field from different levels of backbone. Multi-scale context is contributed from multiple levels of ResNet [21–23], three last features of fully convolution network [24], and mid-level maps [25,26]. On the other hands, receptive fields can obtain from multi-scale inference [27,28].

Feature information fusion and attention mechanism: We can realize the rich contextual information and high resolution by deploying multiple branches, so this approach have great potential in computer vision. Feature fusion are attempted from various studies to continuously enhance the performance. In [29], they focus on solving the semantic gap in different feature maps via matrix multiplication to alleviate contextual loss. Generally, feature fusion is combined by concatenation operation [30] or element-wise summation [27]. These combinations propagate high-level contextual information to the lower level. The hinder appears because targets are calculated without influences between multi-scale features. An attention module is utilized to make connection or guidance between features. Andrew Tao uses a low scale inference to generate an attention mask [28]. This mask contain segmentation characteristics of the low scale, and then it controls and affects the contributions of a high scale inference to the targets. In [16], they employ an attention pyramid and an adaptive fusion to selectively highlight scale weight parameters. The best segmentation of each scale will be contributed to the final. The other approaches exploit a aggregation feature with dynamic attention [31], channel attention [32,33], or self-attention [34].

3. Method

In this section, we first introduce our proposed architecture. Secondly, we describe characteristics of FPN backbone. Lastly, we explain our novel parts: 1) dual prediction, 2) attention module, 3) multi-scale fusion.

3.1. Proposed architecture

Proposed architecture of the feature pyramid network with multi-scale prediction fusion is illustrated in Fig. 2. Our backbone is inspired by the FPN framework [13] and we implement it by

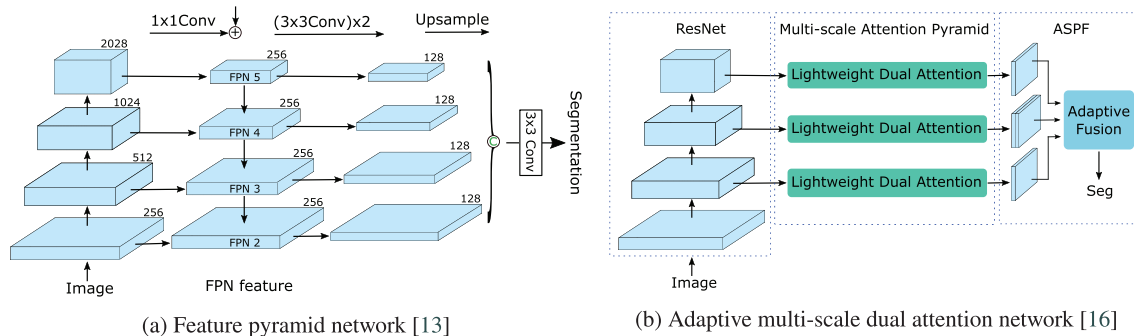


Fig. 1. Illustration of the architectures of previous studies for semantic segmentation. a) baseline method includes a bottom-up pathway and top-down pathway, using average pooling to generate the output. b) a FPN decoder modification is designed with a multi-scale attention pyramid and an adaptive multi-scale prediction fusion (ASPF) modules.

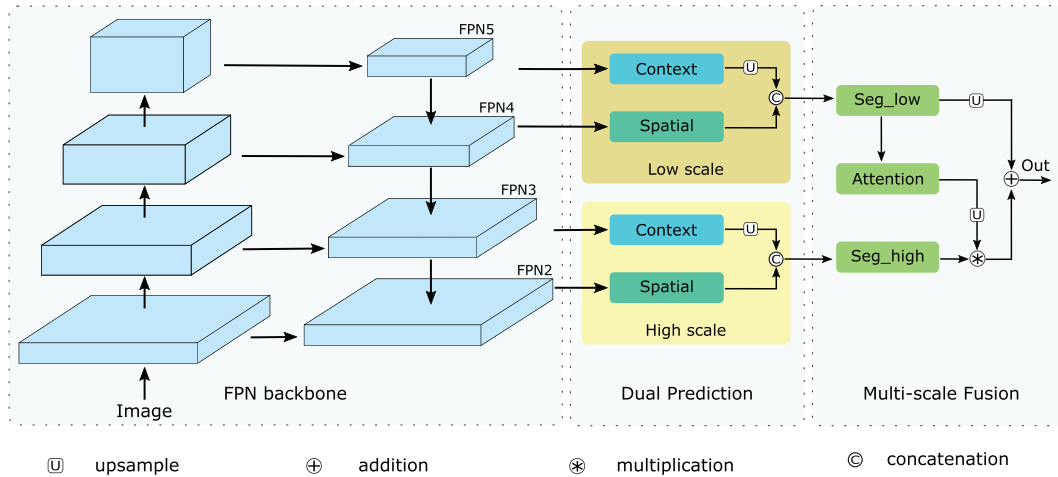


Fig. 2. Proposed architecture for feature pyramid network with multi-scale prediction fusion.

modifying the decoder of the baseline method. The entire architecture includes three main parts.

In the first part, we describe overall the FPN backbone network. We utilize ResNet-50 as the semantic encoder to extract features from input images. The four extracted features are represented in the first column. They are straightly fed into 1x1 convolution to generate new features. Feature pyramid structure is the second feature column which receives information from the ResNet outputs. The FPN consists of four features with different sizes to capture contextual information. The FPN deploys multi-scale features to have the different receptive fields. The detail characteristics of each FPN layer will be analyzed in next section.

Instead of feeding all features into 3×3 convolution and then upsampling the FPN 3, FPN 4, and FPN 5 to the same size of the FPN 2, we design a dual prediction to process the extracted features. It includes one low scale prediction and one high scale prediction. The resolution of a low scale refers to the size of the FPN 4, and a high scale is the FPN 2. By deploying the novel architecture, the first advantage helps reduce the model parameters, and the second is effectively captured the prominent characteristics of each feature pair.

In the third part, multi-scale fusion with attention module is proposed. It is used to combine the outputs of the dual prediction. The low scale can achieve high accuracy for large objects and the high scale can suitably predict small objects. If we combine the outputs by adding together at pixel levels, it leads to average the learnable weights. Therefore, we utilize an attention module between two scales to solve the mentioned drawback.

3.2. Characteristics of baseline FPN network

In the field of deep learning and computer vision, the CNN is utilized to analyze the visual imagery. The CNNs include a series of multiple layers, where the first layer contains basic features with high resolution, and the deeper layers contain rich contextual information. The resolution at deeper layers is rapidly degraded compared to the first layer.

Baseline FPN network contains all characteristics of the CNN layers. We visualize the feature pyramid, predictions, and corresponding ground truths of three samples illustrated in Fig. 3. Firstly, we discuss the detail of different FPN layers. The FPN 5 layer contains rich semantic information with low resolution. The features only have information of complex shapes and almost loss information of objective boundaries. Oppositely, the FPN 2 layer comprises of fine information with high resolution. We realize that the objective boundaries of the FPN 2 layer have high values and

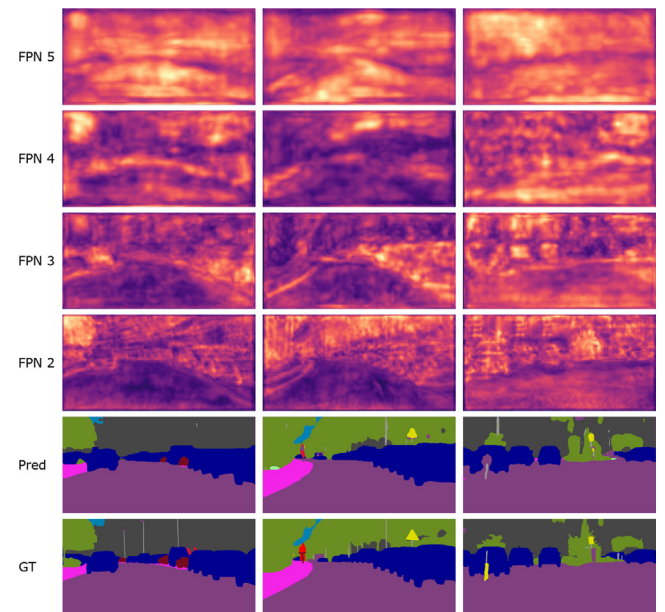


Fig. 3. Illustrate feature characteristics of the baseline FPN at different levels. The heatmaps are generated by model in Fig. 1a. The rows are four-features scales of FPN, prediction masks, and ground truth, top to bottom respectively. The bright color indicate high values, and the dark color is small values.

represent in a brighter color, eg. car boundaries. Two middle layers contain both semantic segmentation and spatial information with different densities. In the baseline study, they improve spatial information of three-top layers by upsampling to the same size of the bottom layer. Thereafter, the feature concatenation is passed through sequential of convolution to generate N_{channels} of classes.

By employing baseline structure, we will encounter two main hindrances. The rich semantic information is reduced when all features are concatenated together. The computation and the number of parameters are increased when three-top layers are upsampled to improve the spatial information. We discuss the detail in the ablation study section.

3.3. Dual predictions

Inspired by the characteristics of the FPN network, we design a new architecture to leverage useful information of each FPN layer.

Four extracted features are separately divided into two branch predictions.

The dual predictions are two distinct branches known as a low scale and a high scale. Depending on the highlighted characters of each feature, we group FPN 5 and FPN 4 layers as the inputs of a low scale branch. The other layers are the inputs of a high scale. In Fig. 4, we describe the architecture of a low scale prediction. The FPN 5 contains richer semantic information, so it mainly contributes context to the next process. To effectively capture the various sizes of objects, the FPN 5 is passed through the atrous spatial pyramid pooling part. The FPN 4 has the function as spatial information provision, and it is simply transferred the features to the concatenation step. The outputs of two features are concatenated together to obtain spatial and contextual information. Finally, feature concatenation is fed into 3x3 convolution to generate the representation of class channels. The architecture for a high scale is similar where the FPN 3 feature is a contextual supplier and the FPN 2 contributes spatial information.

By utilizing dual predictions, they requires less computation than the baseline method. Instead of generating four 128-feature layers and then concatenating features together, the proposed dual directly predicts the representations of class channels.

3.4. Attention module

The mission of semantic segmentation is to assign each pixel to a particular class. The main challenge is the pixel classification at

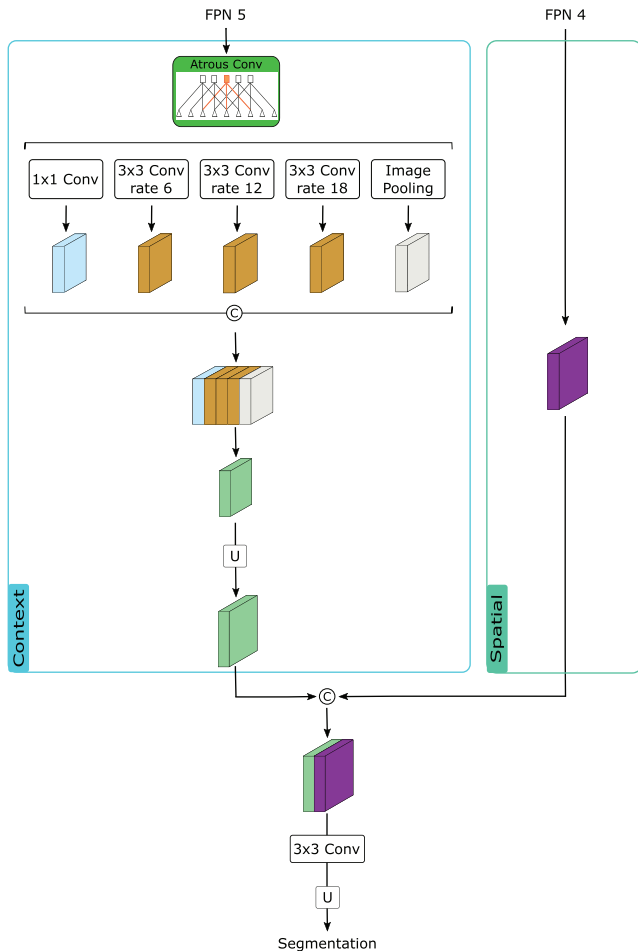


Fig. 4. Architecture of a low scale prediction. A high scale prediction is the same with different input features.

boundaries where it will encounter at least two probabilities of possible classes. Therefore, semantic segmentation generally has good predictions for large objects such as cars, buildings, or vegetation. Poles, fences, and traffic lights have narrow and small structures, so it leads to low precision. To improve the performance, we concentrate on solving the uncertain prediction of small objects and objective boundaries.

In the network, each scale generates separately the segmentation map including N_{class} channels of the datasets. The attention masks are produced by subtracting the low-scale segmentation map by a factor of "1". Therefore, the pixel values of attention channels are strictly dependent on those of the low-scale segmentation. The goal of the attention masks is to discover the uncertain areas or high probabilities of wrong labels. When calculating the feature values with standard convolution, neighborhood pixels are the inputs for a center-pixel computation. Hence, the objective body's pixels have high precision for the prediction. Pixels at the boundaries have the inputs at least from two different classes. To impress the attention function, we visualize information for the explanation. Fig. 5 includes images, labels, prediction masks, and attention masks. Prediction masks are generated from a top layer of the low scale branch. The light green color of attention masks indicates uncertain prediction or high values, and the dark blue color is low values. We draw the red circles to illustrate a function of the attention module. When the outputs of predictions are uncertain to correctly match to the class labels, the attention masks will have high values and are denoted in green color. Fig. 5 shows that small objects and objective boundaries are the most uncertain areas. In this section, we simply describe the function of the attention module on the low scale segmentation. This attention also affects the high scale segmentation, and it will be explained in Section 3.5.

The attention mask is calculated as Eq. 1:

$$S_A = \sum_{i=0}^C \sum_{j=0}^{H+W} (1 - S_{lo_{ij}}) \quad (1)$$

where S_A is an attention masks and S_{lo} is the low scale mask. H, W , are height, width of the low scale and C is numbers of class channels. " i, j " $\in \mathbb{R}$ and denote pixel locations.

3.5. Multi-scale fusion

We make a prediction of the feature fusion part to illustrate the advantages and disadvantages of each scale. As the above explanation, a low scale segments good results with large objects and on the near screen. The segmentation of such cars and road has high

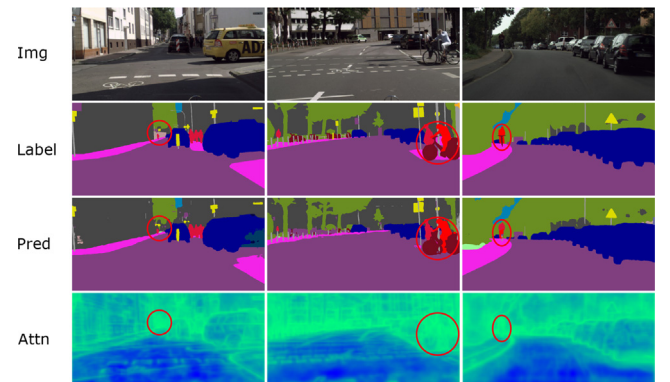


Fig. 5. Illustrate attention masks. The rows are images, labels, prediction masks and attention masks of the low scale, top to bottom respectively. The light green color of attention masks indicates uncertain prediction or high probabilities of wrong labels.

accuracy, but it is bad to predict the small objects. Inversely, a high scale having small receptive fields can capture and predict small objects with high precision. However, the hinder of the small receptive fields is not covered completely the whole contextual information of the large object, especially large objects near the screen. Instead of deploying the segmentation fusion with a feature addition or using max-pooling methods, we utilize an attention module between two segmentation maps shown in Fig. 6. The attention mask is generated from a low segmentation and its characteristics are explained detail in Section 3.4. The first function of the attention module is discovering the high probabilities of wrong labels. The light green color is uncertain areas and locates at objective boundaries and on the far screen. The second function of this module is the effects on the high scale. The high scale prediction is multiplied by the attention channels to generate new learnable weights of this branch, and the formula is expressed in Eq. 2. By this way, the attention module directly controls the contribution of a high scale. It not only selects prominent weights but also eliminates bad predictions before providing to the final fusion.

The high scale attention is calculated as follows:

$$S_{A_{hi}} = \sum_{i=0}^C \sum_{j=0}^{H+W} (S_{A_{ij}} * S_{hi_{ij}}) \quad (2)$$

In our approach, the segmentation output is effectively combined from two scale maps. When a low scale provides completely learnable weights to the final, a high scale only complements the bad prediction of a low scale. Small objects and objective boundaries are usually uncertain predictions, so the weight outputs of these areas are mainly contributed by both scales. The final segmentation weights can be expressed as Eq. 3.

$$\begin{aligned} S &= S_{lo} + S_{A_{hi}} \\ \text{or} \\ S &= [\mathcal{U}(S_{lo})] + [\mathcal{U}(1 - S_{lo}) * S_{hi}] \end{aligned} \quad (3)$$

where S is the semantic segmentation output. Three main components are a low scale segmentation S_{lo} , a high scale segmentation S_{hi} , and an attention mask S_A . Lastly, $\mathcal{U}$ is the bilinear upsampling operation.

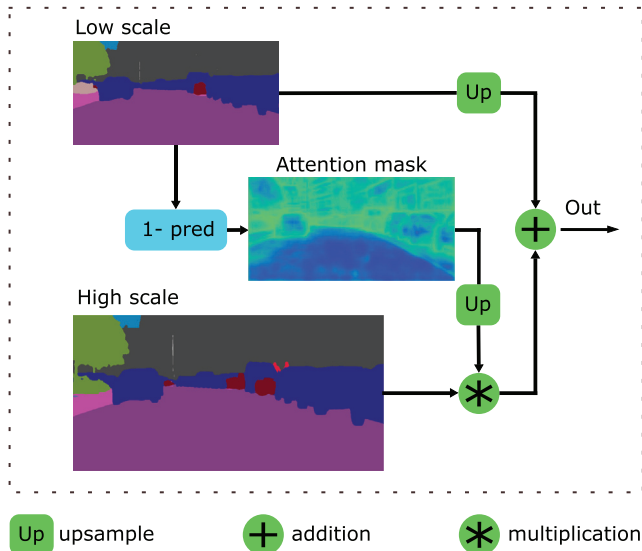


Fig. 6. Multi-scale fusion with an attention module.

4. Experiments

4.1. Implementation details

In the experiment section, we implement our study on the Pytorch framework and set up the initial parameters for the training model such as 150 epochs, a batch size of 2 images per GPU, weight decay as 0.0005, and momentum of 0.9. A poly learning rate policy is utilized during the training process where an initial learning value is 0.02. The mini-batch stochastic gradient descent (SGD) is the algorithm for the optimizer. We use the cross-entropy function to calculate loss, and intersection of union (IoU) to evaluate the performance of the proposed approach. For cityscapes dataset evaluation, we conducted experiments on Nvidia Titan X GPU with 12 GB of GDDR5X memory. On the other hand, a GeForce RTX 3090 having 24 GB and GDDR6X is used to train our module on mapillary vistas dataset.

The cross-entropy loss is shown as following equation:

$$L = \frac{1}{N} \sum_{i=1}^N (-y_i \log(p_i)) \quad (4)$$

where L is the loss value and N is the size of each dataset. The calculation between true labels y_i and the model predicted probability p_i .

4.2. Ablation study for proposed components

In this section, we conduct a series of experiments to validate the effectiveness of each proposed component. Compared to the baseline FPN method, our architecture employs the novel dual prediction and the multi-scale fusion. Table 1 describes the ablation study for proposed components with respect to the number of model parameters and mIoU. The baseline method achieves 71.3% mIoU with 31.7 M parameters. When we modify the baseline by using the dual prediction module, the network increases to 76.1% mIoU and especially reduces 0.9 M parameters. The number of model parameters is reduced because the low-scale prediction has processed the FPN 5 and FPN 4 layers in lower resolutions than the baseline network. When two proposed components are used, the performance is improved 1.8% mIoU and maintains the same parameters compared to only one dual prediction being used. Conclusively, our proposed approach not only improves significantly 6.6% mIoU but also reduces the computation for the network.

4.3. Ablation study for dual prediction

In order to particularly impress our novel dual prediction, we demonstrate the impact of each branch shown in Table 2. Except for the baseline method, the multi-scale fusion module is applied in all experiments. The core things of the dual prediction are an atrous spatial pyramid pooling (ASPP) of the contextual branch and the contribution of the spatial branch. All experiments achieve better performance and require less computation than the baseline method. In the case of the spatial elimination, the parameter numbers are significantly reduced to 30.3 M and the accuracy still reaches to 72.6% mIoU. However, when the dual prediction is modified in both cases, the accuracy only surpasses the FPN method approximately 1.3% mIoU. The results show that the dual prediction module critically improves the model performance, and they can account for 4.8% mIoU of total 6.6% accomplishment of our proposed method.

Table 1

Ablation study for proposed components on Cityscapes validation set. DP denotes dual prediction module and MSF represents the multi-scale fusion module.

Method	BaseNet	DP	MSF	No. params	mIoU (%)
FPN [13]	ResNet50			31.7 M	71.3
FPN + DP	ResNet50	✓		30.9 M	76.1
Ours	ResNet50	✓	✓	30.9 M	77.9

Table 2

Ablation study for dual prediction on Cityscapes validation set. ASPP is the atrous spatial pyramid pooling of context path. Spatial is spatial path in dual prediction modules, and MSF represents the multi-scale fusion.

Method	BaseNet	ASPP	Spatial	MSF	No. params	mIoU (%)
FPN [13]	ResNet50				31.7 M	71.3
FPN + Spatial + MSF	ResNet50		✓	✓	30.9 M	72.3
FPN + ASPP + MSF	ResNet50	✓		✓	30.3 M	72.6
Ours	ResNet50	✓	✓	✓	30.9 M	77.9

Table 3

Ablation study for multi-scale fusion on Cityscapes validation set. Concatenate and addition are the concatenate pooling and addition function, respectively. Attention represents an attention module of the proposed multi-scale fusion.

Method	BaseNet	DP	Concatenation	Addition	Attention	No. params	mIoU (%)
FPN [13]	ResNet50					31.7 M	71.3
FPN + DP + Concatenate	ResNet50	✓	✓			32.1 M	73.4
FPN + DP + Addition	ResNet50	✓		✓		30.9 M	76.1
Ours	ResNet50	✓			✓	30.9 M	77.9

Table 4

Accuracy comparison for object categories on the Cityscapes. The classes are road, side walk, building, wall, fence, pole, traffic light, traffic sign, vegetation, terrain, sky, person, rider, car, truck, bus, train, motorcycle, and bicycle, left to right respectively.

Method	Road	swalk	build	wall	fence	pole	tlght	tsign	veg.	terr	sky	pers.	rider	car	truck	bus	train	mcle	bicle	mIoU
ESPNet[3]	97.3	78.6	88.8	43.5	42.1	49.3	52.6	60.0	90.5	66.8	93.3	72.9	53.1	91.8	53.0	65.9	53.2	44.2	59.9	66.2
CGNet[35]	97.7	81.0	89.8	42.5	48.0	56.2	59.8	65.3	91.4	68.2	94.2	76.8	57.1	92.8	50.8	60.1	51.8	47.3	61.7	68.0
EDANet[36]	97.8	80.6	89.5	42.0	46.0	52.3	59.8	65.0	91.4	68.7	93.6	75.7	54.3	92.4	40.9	58.7	56.0	50.4	64.0	67.3
CFPNNet[37]	97.8	81.4	90.5	46.4	50.6	56.4	61.5	67.7	92.1	68.9	94.3	80.4	60.7	93.9	51.4	68.0	50.8	51.2	67.7	70.1
RELAXNet[4]	98.9	84.9	92.2	57.2	54.8	64.3	70.6	74.0	93.0	71.8	94.8	83.7	64.4	95.1	58.6	72.7	58.2	59.9	71.8	74.8
DSANet[38]	96.8	78.5	91.2	50.5	50.8	59.4	64.0	71.7	92.6	70.0	94.5	81.8	61.9	92.9	56.1	75.6	50.6	50.9	66.8	71.4
FANet[39]	97.9	83.3	91.6	55.5	55.1	60.3	66.2	74.9	91.7	61.8	94.7	78.5	58.1	94.1	76.8	85.1	74.5	50.7	73.9	75.0
AGLNet[40]	97.8	81.0	91.0	51.3	50.6	58.3	63.0	68.5	92.3	71.3	94.2	80.1	59.6	93.8	48.4	68.1	42.1	52.4	67.8	70.1
FDDWNet[41]	98.0	82.4	91.1	52.5	51.2	59.9	64.4	68.9	92.5	70.3	94.4	80.8	59.8	94.0	56.5	68.9	48.6	55.7	67.7	71.5
RefineNet[42]	97.9	81.3	90.3	48.8	47.4	49.6	57.9	67.3	91.9	69.4	94.2	79.8	59.8	93.7	56.5	67.5	57.5	57.7	68.8	73.6
FPN[13]	97.5	81.6	90.9	46.3	54.2	59.1	63.9	74.3	91.3	58.8	93.6	78.4	56.0	93.3	54.0	71.9	54.7	59.6	74.8	71.3
Ours	98.1	85.1	92.6	56.3	60.7	65.1	67.9	76.7	92.5	65.5	94.9	80.6	60.3	94.8	81.4	86.8	79.6	64.2	76.3	77.9

4.4. Ablation study for multi-scale fusion

Another aspect of the ablation study is the multi-scale fusion. In these experiments, we only focus on different approaches to combine the two predictions as shown in Table 3. The concatenation operation slightly enhances the accuracy by 2.1% mIoU, but it also accumulates the model parameters to 32.1 M. When the outputs of dual prediction are concatenated together, they are fed into convolution layers to generate the presentation of class channels, so these cause to increase the computation. The addition approach achieves a high accuracy 76.1% mIoU with 30.9 M parameters, and this method surpasses the concatenation study by 2.7 mIoU. Finally, the multi-scale fusion utilizes our attention module. The proposal is seemly to have the same number of parameters of the addition combination, but it improves the performance to 77.9% mIoU.

4.5. Results on cityscapes

Cityscapes is a common dataset for semantic segmentation which is captured from 50 different countries. It contains 19 objec-

Table 5

Comparison of our method and other SOTA existing methods in terms of accuracy and speed on Cityscapes

Method	Resolution	mIoU (%)	FPS
ICNet [43]	1024 × 2048	67.7	38
BiseNet [10]	768 × 1536	74.8	47
SwiftNet [44]	1024 × 2048	75.4	40
AGLNet [40]	512 × 1024	70.1	52
SynNet [45]	1024 × 2048	76.3	50
FANet [39]	1024 × 2048	75.0	72
ADANet [16]	1024 × 2048	77.3	15
FPN [13]	1024 × 2048	71.3	18
Ours	1024 × 2048	77.9	62

tive classes in the urban streets. The whole dataset has 5,000 images and is divided into three sets such as 2,975 images for the training set, 500 images for the validation set, and 1,525 images for the test set. All images have a resolution of 2048 × 1024 pixels.

In terms of particular class accuracy, our method mostly has the highest values described in Table 4. Uncountable objects such as

road, vegetation, or sky achieve accuracy higher than 90% mIoU for all approaches. To impress the influence of our method, we concentrate on the low output precision. When other existing methods only obtain the accuracy approximately 50% mIoU for truck and train classes, we have a great improvement with 81.4% and 79.6%, respectively. Our results also demonstrate the enhancement of narrow objective structures including fences, poles, or traffic lights. Compared to the baseline method FPN [13], we surpass in all aspects of comparison. Conclusively, our approach achieves high accuracy and has good results for both large and thin objects.

Table 6
Comparison of our method and other SOTA existing methods in terms of accuracy on Mapillary Vistas.

Method	Resolution	mIoU (%)
AGLNet [40]	1024 × 2048	30.7
DABNet [2]	1024 × 2048	29.6
RGPNet [46]	1024 × 2048	41.7
ADANet [16]	1024 × 2048	42.5
FPN [13]	1024 × 2048	40.2
Ours	1024 × 2048	44.1

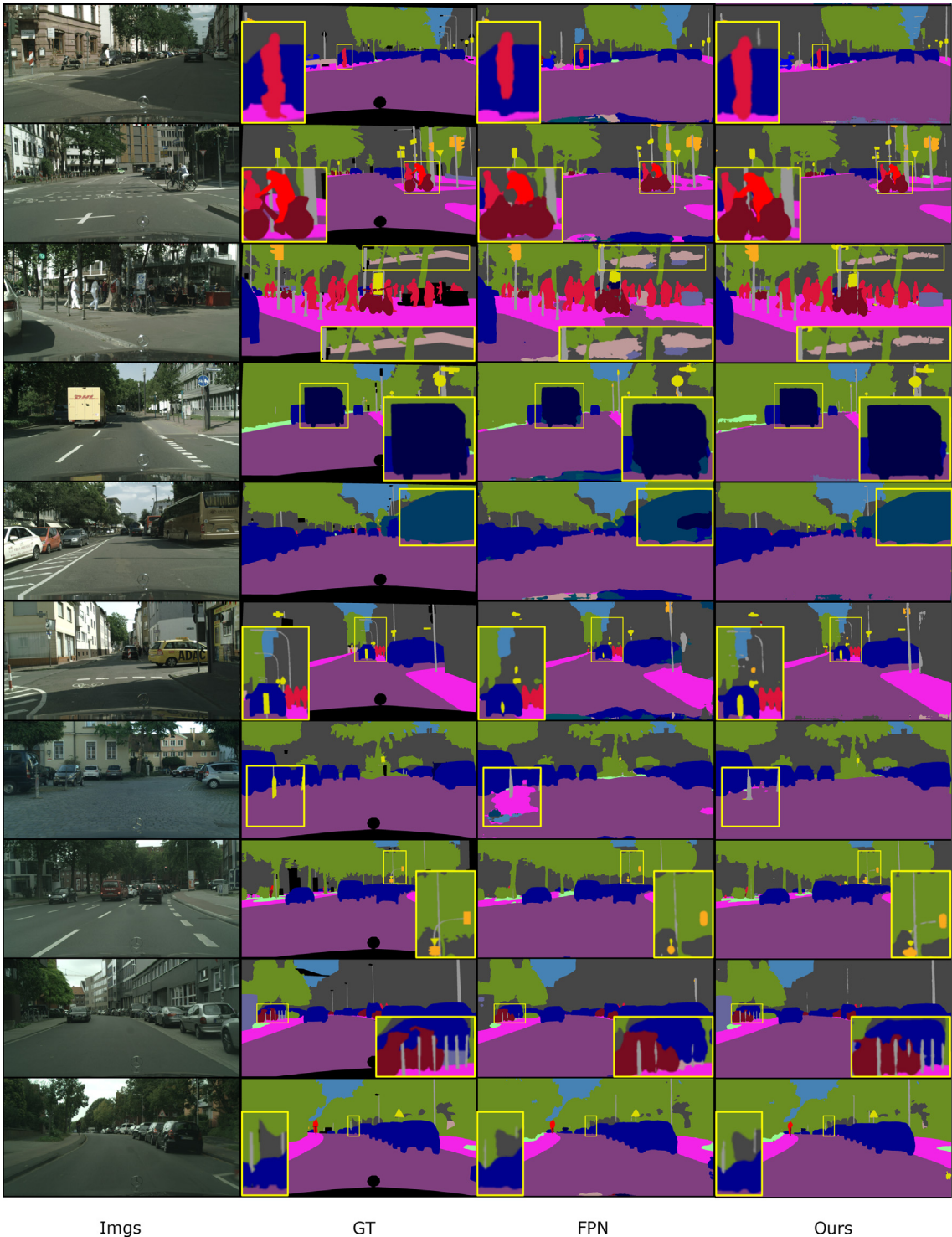


Fig. 7. Qualitative results of the baseline FPN and our proposal approach on Cityscapes.

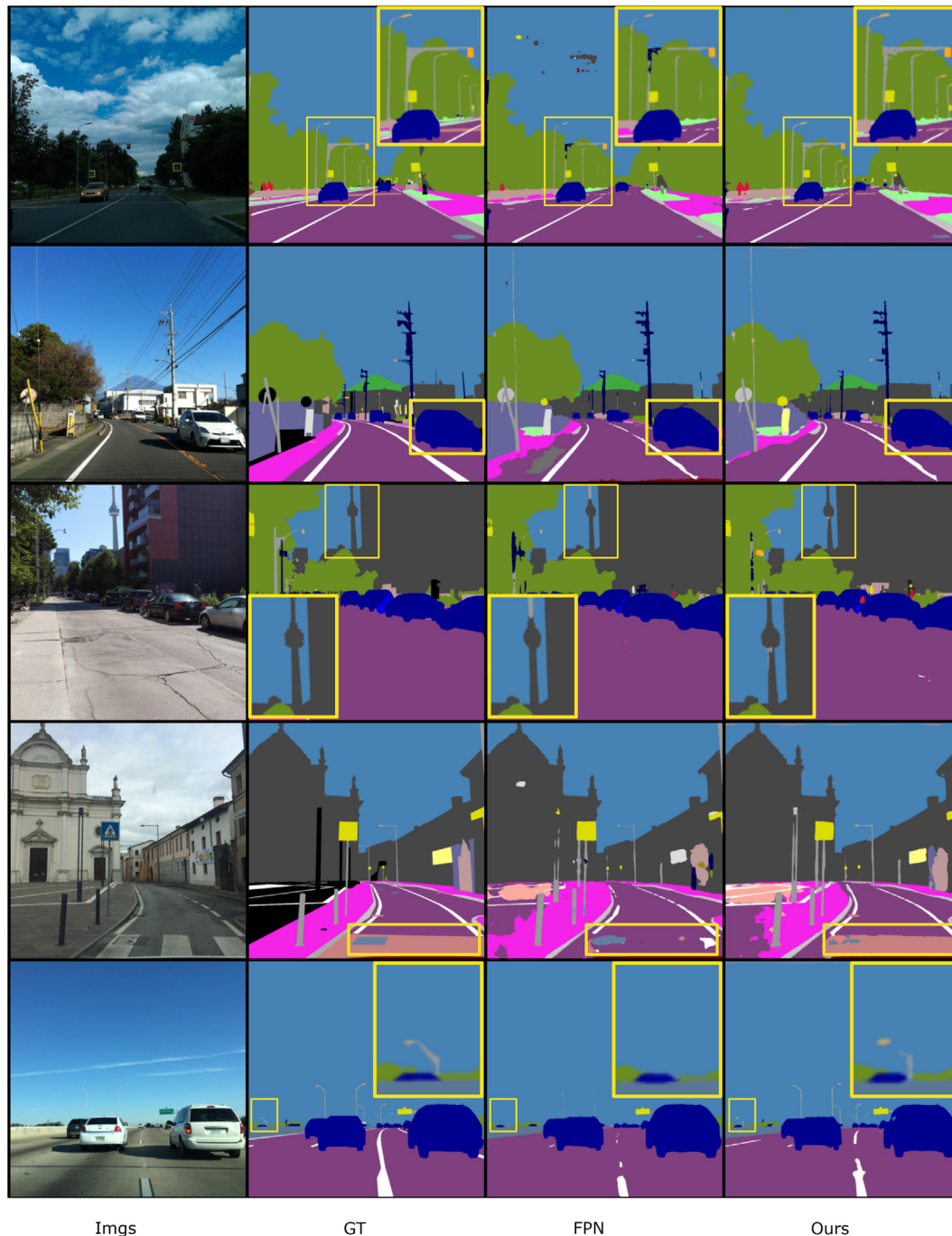


Fig. 8. Qualitative results of the baseline FPN and our proposal approach on Mapillary Vistas.

In Table 5, we compare our proposed method to other existing methods on the facets of performance and speed. Our results dominate the baseline FPN method [13] with 6.6% mIoU improvement and three times faster. We also outperform the other algorithms with respect to the performance. Our results exceed ICNet approach [43] by 10.2%. When ADANet method [16] reaches the highest accuracy, we are still better than them 0.6% mIoU, and especially they only has 15 FPS in term of the speed. On the other hand, the inference speed of our model is crucially improved and achieves real-time segmentation. Only the FANet method has less time consumption than our proposal, but its accuracy is lower than ours by 2.9% mIoU. The results show that our model has a good

trade-off between time consumption and accuracy for semantic segmentation.

The qualitative results of our proposal and the baseline FPN network are visualized in Fig. 7. We can categorize objective classes into large, medium, and small sizes to analyze the visualization. Medium-size cars always have good segmentation for both approaches. The narrow structures consist of people, traffic lights, or poles, so our method identifies them more correctly than the baseline method. For large objects such as trucks or roads, the FPN method can not capture the whole information and then leads to wrong label classes. Our attention module refines the contribution of a high scale, so it neglects or elimi-

notes the bad prediction of large objects visualized in rows 4 and 5 of Fig. 7.

4.6. Results on Mapillary vistas

Mapillary includes diverse datasets for computer vision applications. In our semantic segmentation study, we use Mapillary Vistas dataset which is pixel-accurate annotation for understanding street scenes. Data is collected on the worldwide streets and has 65 semantic object categories. Whole 20,000 images split into a training set of 18,000 images and 2,000 images of a validation set.

To verify the impact of our approach, we evaluate the model on a different dataset. Mapillary Vistas is common known as large and complex data with 65 objective classes. It includes some difficult-recognition objects such as birds, ground animals, or CCTV camera. Our method achieves 44.1% mIoU for segmentation performance shown in Table 6. We outperform the other existing approaches, and especially we surpass the DABNet [2] with 14.5%. Compared to the baseline network, we are also better than them 3.9%. The ADANet [16] achieves the high accuracy 42.5% mIoU, but we have exceeded them 1.6%.

Finally, qualitative results of our proposal and the baseline are visualized in Fig. 8. Although the mIoU has low values, the segmentation of Mapillary is clearly visualized for both narrow and large objects in the streets such as traffic lights, cars, or roads. Our method is more effectively captured thin and large objects than the FPN approach.

5. Conclusion

Based on the characteristics of FPN layers, we optimize the feature pyramid network structure to improve semantic segmentation tasks. The dual prediction is designed to capture information with different receptive fields, so it can efficiently segment both narrow and large objects in complex street scenes. Moreover, the novel dual not only significantly improve the performance but also reduce the parameter numbers of the model. The multi-scale fusion is employed to select learnable weights from different scale maps. The proposed attention module of the fusion part discovers uncertain prediction of a low scale, and then it controls the contribution of a high scale for the complementation. The accuracy results illustrate the reasonable balance between thin and large objects. This study improves segmentation performance while still achieving real-time application. In the future, we work on the extracted network to strengthen the semantic accuracy. The proposed FPN structure can be adjusted to suitably apply to numerous vision application, especially on the object detection field.

CRediT authorship contribution statement

Toan Van Quyen: Conceptualization, Methodology, Software, Visualization, Writing – original draft. **Min Young Kim:** Supervision, Methodology, Conceptualization, Funding acquisition, Project administration, Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Min Young Kim reports financial support was provided by Korea Ministry of Education. Min Young Kim reports financial support was provided by National Research Foundation of Korea.

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